

The sine and cosine functions and vertical dilations



1 Consider the expression $\cos \theta$.

a Complete the table of values for different values of θ .

θ	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos \theta$									

b Sketch a graph of the function $y = \cos \theta$.

c State the maximum value of $\cos \theta$.

d State the minimum value of $\cos \theta$.

e State the range of values of $4 \cos \theta$.

2 Consider the functions $y = \sin x$ and $y = \cos x$, where x is in radians.

a State the domain of both functions.

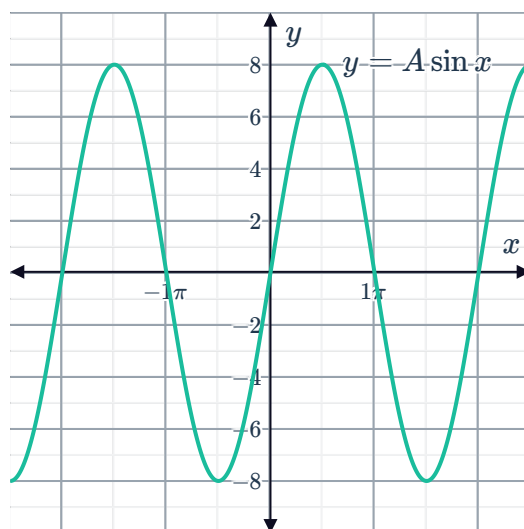
b State the range of both functions.

3 Consider the function $y = -2 \cos x$, where x is in radians.

a State the domain of the function.

b State the range of the function.

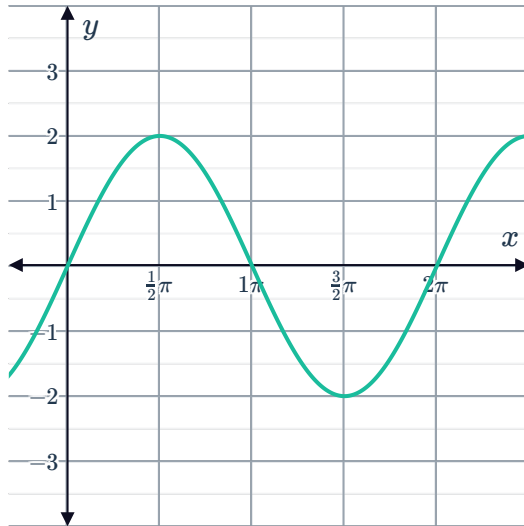
4 Consider the given graph of a function of the form $f(x) = A \sin x$:
State the amplitude of the function.



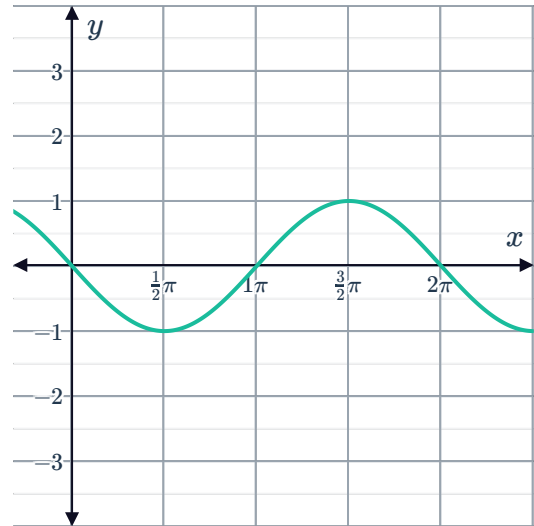
- 5 State the equation of each of the functions plotted below given that they are of the form $y = a \sin x$ or $y = a \cos x$:



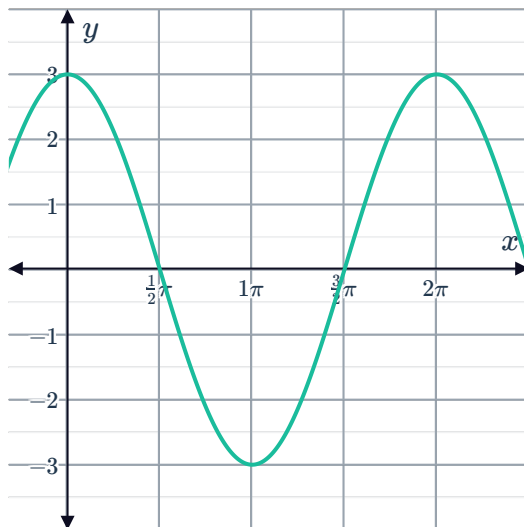
a



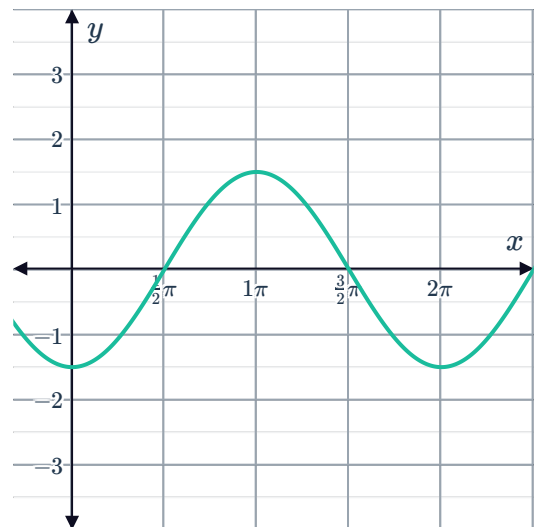
b



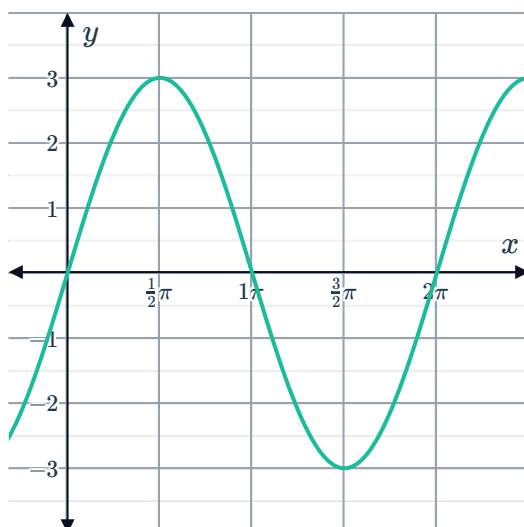
c



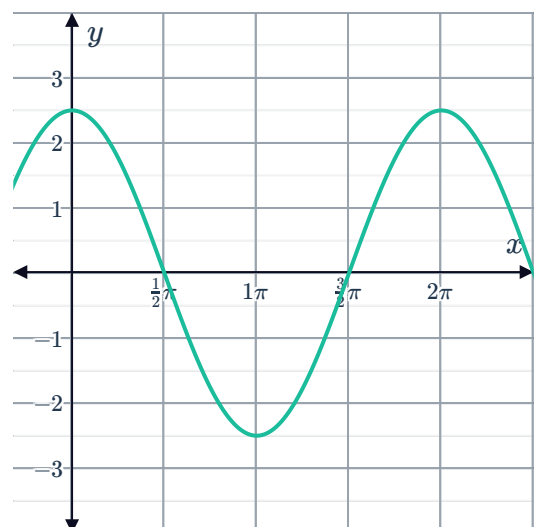
d



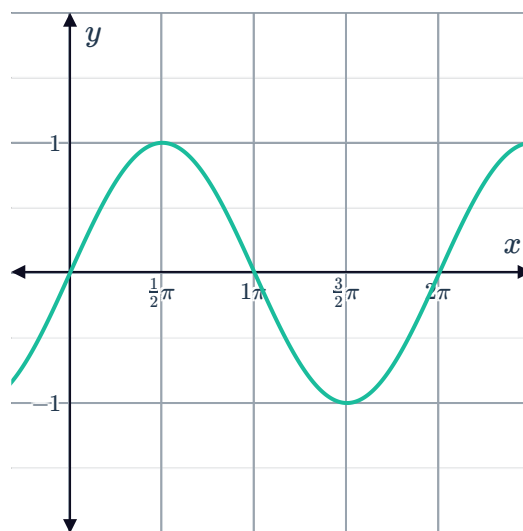
e



f



- 6 Consider the graph of the function $y = \sin x$ for $0 \leq x < 2\pi$:
At which value of x in the given domain would $y = -\sin x$ have a maximum value?

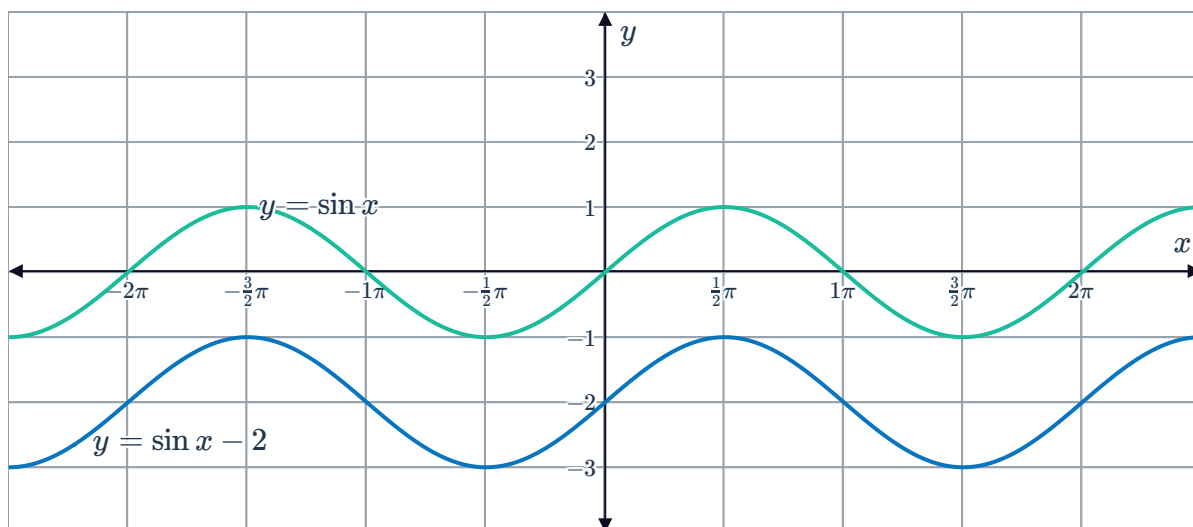


- 7 Consider the function $f(x) = -8 \sin x$, where $0 \leq x \leq \pi$.
- State the amplitude of the function.
 - Find the value of $f(\pi)$.
 - Find the minimum value of the function.
- 8 The function $y = k \sin x$ has a maximum value of 5. Find the value of k , where $k > 0$.
- 9 A sine function has the form $y = c \sin x$, a range of $[-2, 2]$ and a minimum at $\frac{\pi}{2}$. Find an expression for y .
- 10 For each of following functions:
- State the amplitude.
 - Sketch a graph of the function.
- $y = 3 \sin x$
 - $y = 4 \cos x$
 - $y = \frac{4}{3} \cos x$
 - $y = -5 \cos x$
 - $y = \frac{5}{4} \sin x$
 - $y = -4 \sin x$
- 11 Consider the function $y = -3 \cos x$.
- State the maximum value of the function.
 - State the minimum value of the function.
 - State the amplitude of the function.
 - Describe the two transformations required to obtain the graph of $y = -3 \cos x$ from the graph of $y = \cos x$.



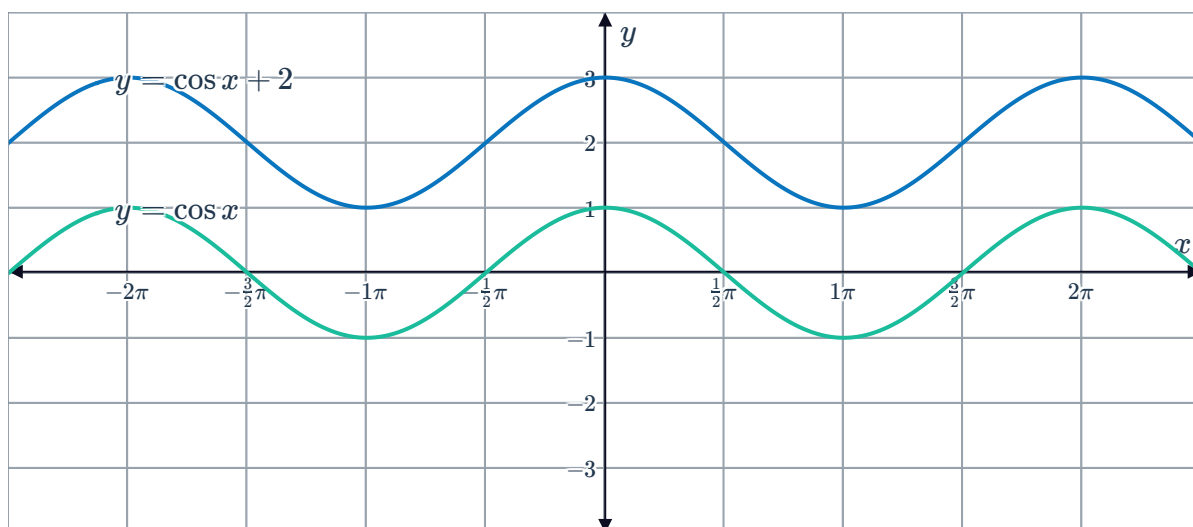
Vertical translations of sine and cosine functions

12 Consider the two graphs $y = \sin x$ and $y = \sin x - 2$ below:



Describe the transformation required to obtain the graph of $y = \sin x - 2$ from $y = \sin x$.

13 Consider the two graphs $y = \cos x$ and $y = \cos x + 2$ below:



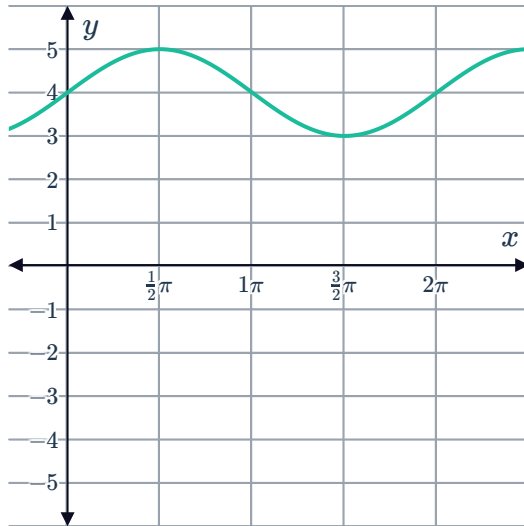
Describe the transformation required to obtain the graph of $y = \cos x + 2$ from $y = \cos x$.

14 Describe the transformation required to obtain the graph of $y = \sin x + 4$ from $y = \sin x$.

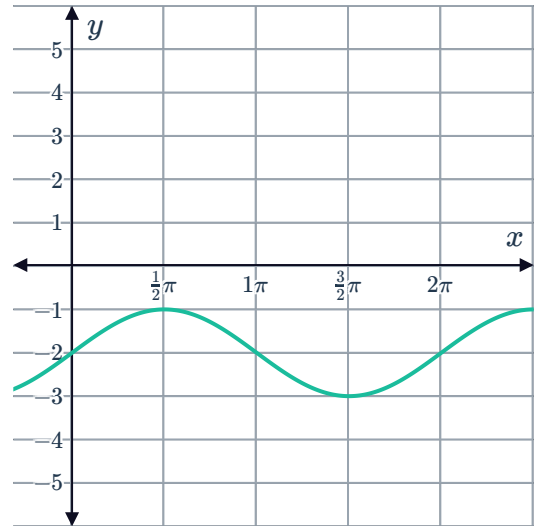
- 15 State the equation of each of the functions plotted below given that they are of the form $y = \sin x + k$:



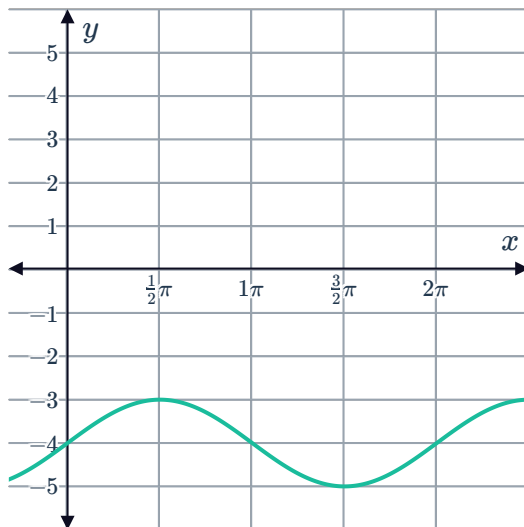
a



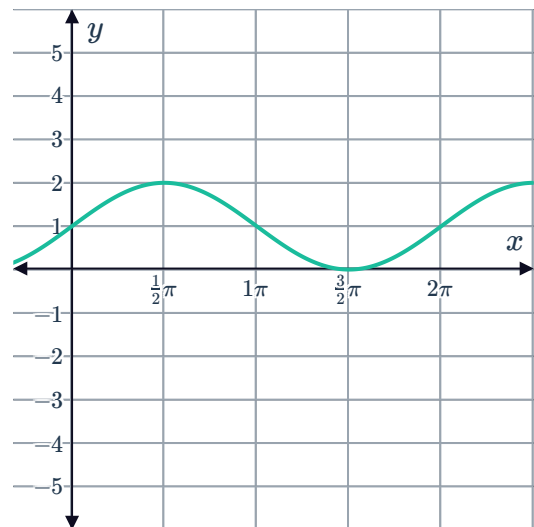
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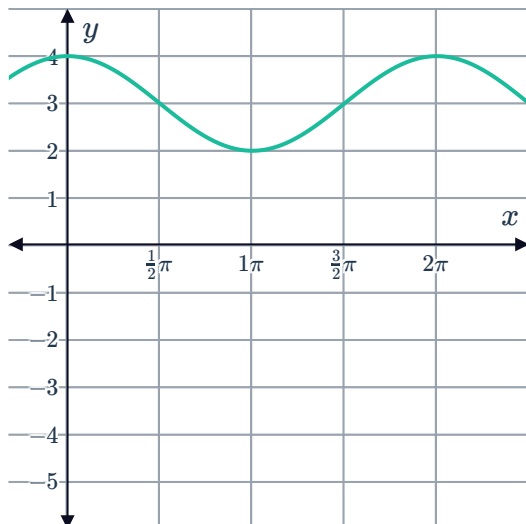


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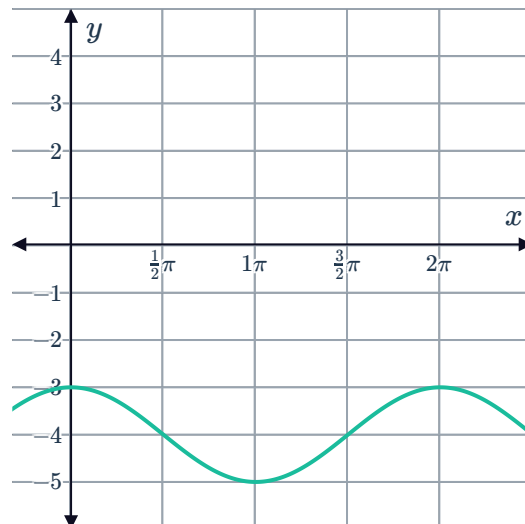


- 16 State the equation of each of the functions sketched below given that they are of the form $y = \cos x + k$:

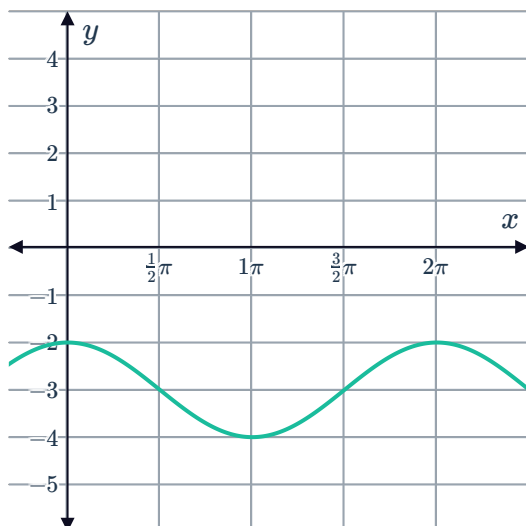
a



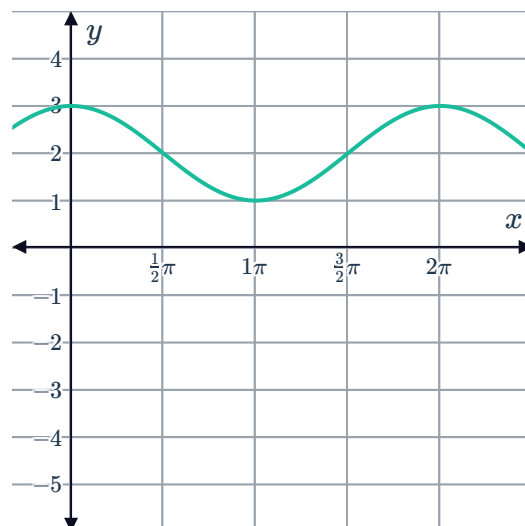
b



c



d



- 17 Describe the two transformations required to turn the graph of $y = \cos x$ into the graph of $y = -\cos x + 3$.

- 18 State the equation of the line that the function $y = \sin x + 2$ oscillates about.

- 19 Consider the function $y = \cos x + 4$.

- Describe the transformation required to obtain the graph of $y = \cos x + 4$ from $y = \cos x$.
- State the period of the function, giving your answer in degrees.
- Find the maximum value of the function.
- Find the minimum value of the function.

20 For each of the functions below:



- i State the period of the function, giving your answer in radians.
- ii State the amplitude of the function.
- iii Find the maximum value of the function.
- iv Find the minimum value of the function.
- v Sketch a graph of the function for $-2\pi \leq x \leq 2\pi$.

a $y = \sin x + 1$ **b** $y = \cos x - 3$ **c** $y = \sin x - 2$ **d** $y = 3 \sin x + 2$
e $y = 3 \cos x - 3$ **f** $y = 2 \sin x - 3$

21 For each of the functions below:

- i Find the value of y when $x = \frac{\pi}{2}$.
- ii State the amplitude of the function.
- iii Find the maximum value of the function.
- iv Find the minimum value of the function.
- v Describe the transformations required to obtain the graph of the function from $y = \cos x$.
- vi Sketch a graph of the function for $0 \leq x \leq 2\pi$.

a $y = 2 \cos x + 3$ **b** $y = 2 \cos x - 3$ **c** $y = -4 \cos x + 2$ **d** $y = \frac{1}{3} \sin x + 2$

22 The function $y = \cos x + 5$ is translated 4 units up.

- a** State the equation of the new function after the translation.
- b** Find the maximum value of the new function.

23 For each of the functions below:

- i Find the maximum value of the function.
- ii Find the minimum value of the function.

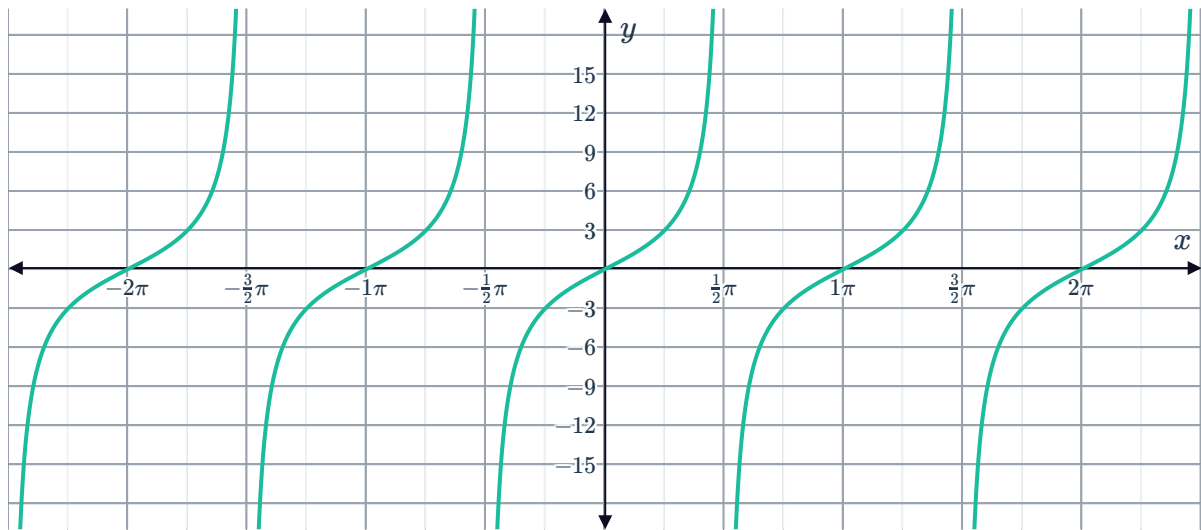
a $y = 2 - 3 \sin x$ **b** $y = 4 - 3 \sin x$ **c** $y = (\sin x)^2 + 1$ **d** $y = \frac{4}{3 \cos x + 5}$

24 A sine function, y , has the form $y = c \sin x + d$ and a range of $[0, 4]$. Find an expression for y , where $c > 0$.

25 A cosine function, y , has the form $y = c \cos x - d$ and a range of $[-10, 6]$. Find an expression for y , where $c > 0$.

The tangent function and vertical dilations

26 Consider the given graph of $f(x) = 3 \tan x$:

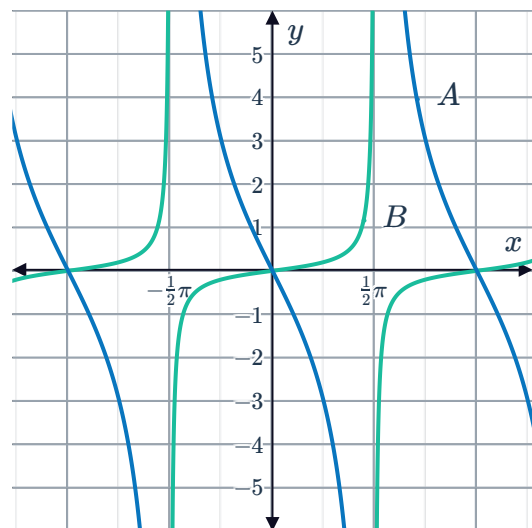


- State the domain of $f(x)$.
- State the range of $f(x)$.
- The graph of $f(x)$ has its domain restricted to $[0, \pi)$, state the range of the restricted graph.

27 Consider the functions $f(x) = \frac{1}{5} \tan x$ and $g(x) = -3 \tan x$ drawn on the given set of axes:

Which function has been labelled:

- A
- B



28 For each of the functions below:

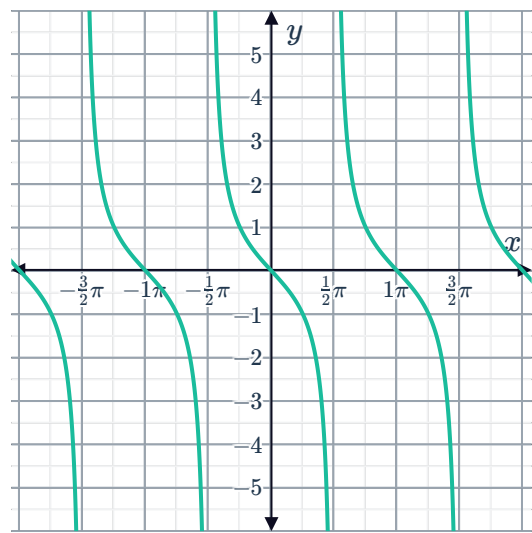
- State the vertical dilation factor required to obtain the given function from $y = \tan x$.
- State whether the graph would appear stretched or compressed in relation to $y = \tan x$.

- $y = \frac{1}{4} \tan x$
- $y = 6 \tan x$
- $y = \frac{5}{3} \tan x$
- $y = 0.8 \tan x$

29 Consider the given graph of a function of the form $y = a \tan x$:

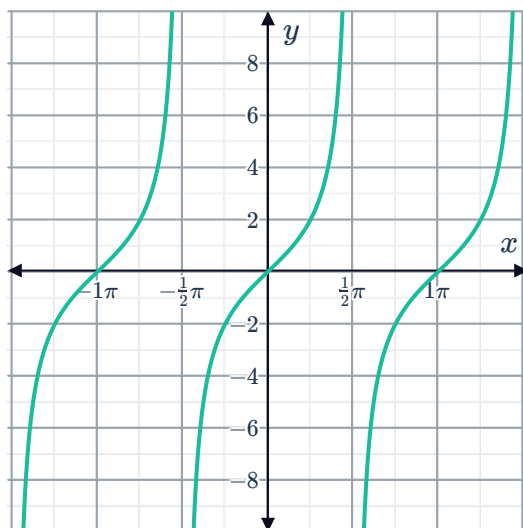


- Describe the transformation required to obtain the given function from $y = \tan x$.
- State the equation of the graphed function.

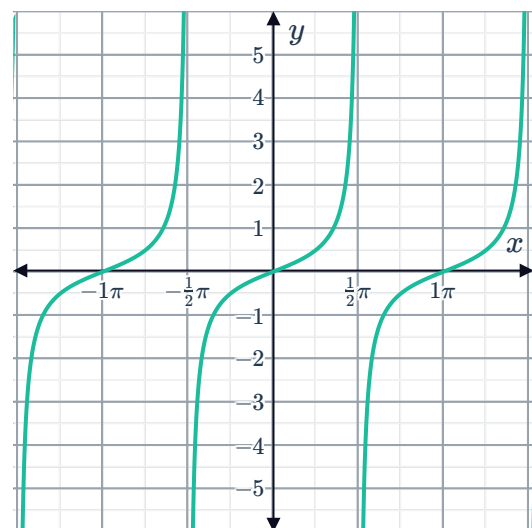


30 State the equation of each of the functions graphed below given that they are of the form $y = a \tan x$:

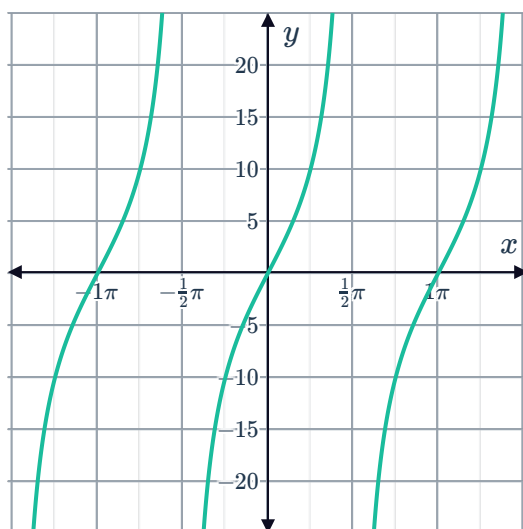
a



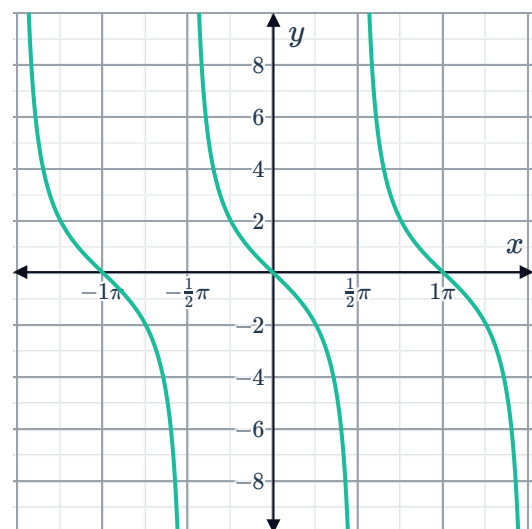
b



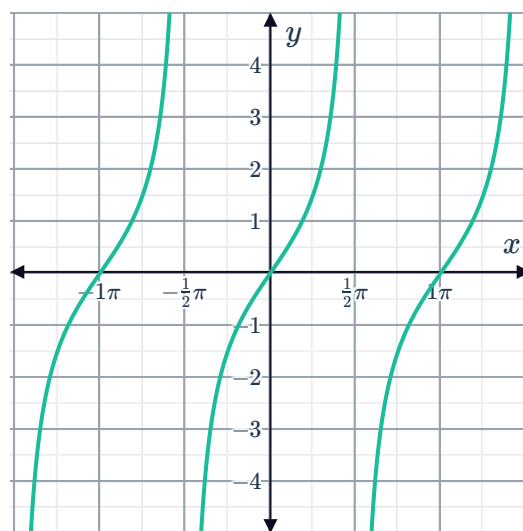
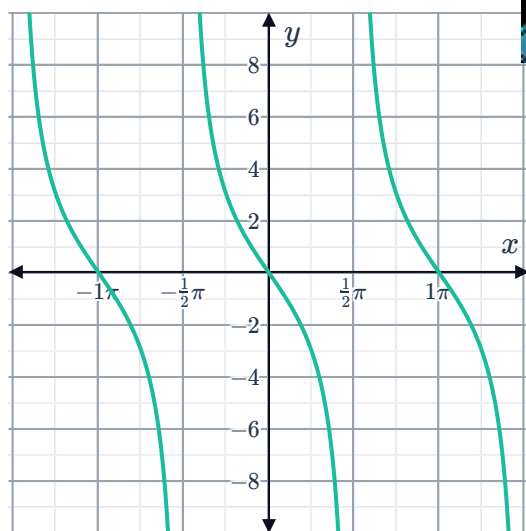
c



d



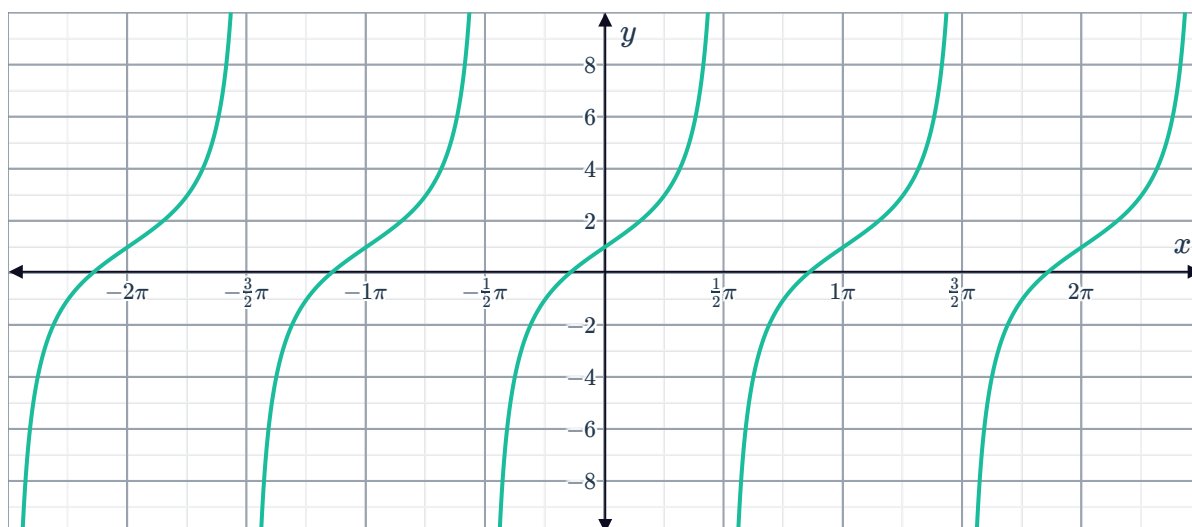
e



- 31 On the same set of axes, sketch the graph of $y = \tan x$ and $y = \frac{1}{2} \tan x$.

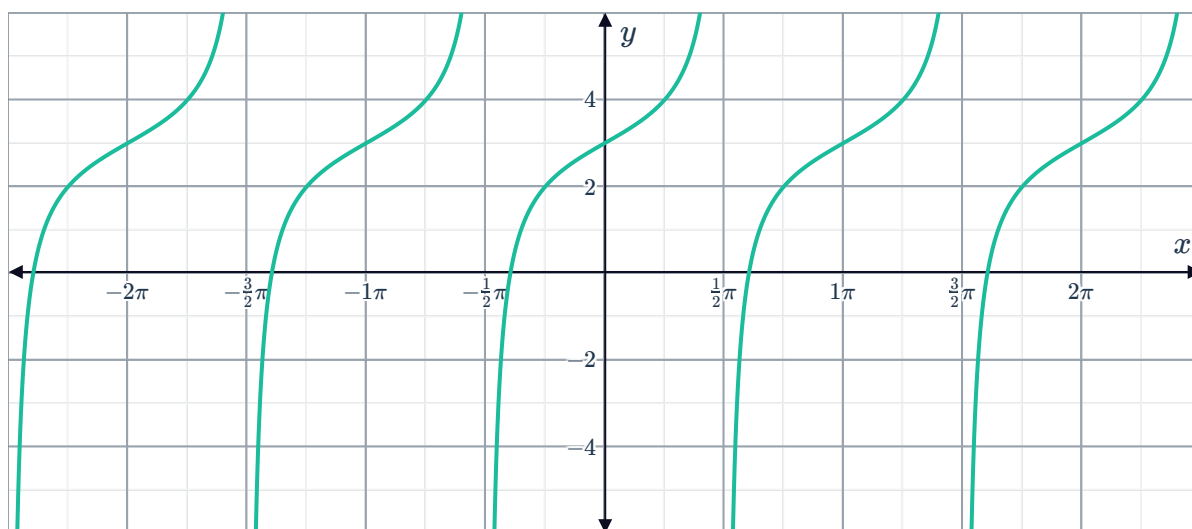
Vertical translations of the tangent function

- 32 Consider the given graph of $f(x) = 2 \tan x + 1$:



- a State the domain of $f(x)$. b State the range of $f(x)$.

33 Consider the given graph of $f(x) = \tan x$



- a** State the domain of $f(x)$.
- b** State the range of $f(x)$.
- c** The graph of $f(x)$ has its domain restricted to $\left[0, \frac{\pi}{2}\right)$, state the range of the restricted graph.

34 For each of the functions below:

- i** Find the y -intercept.
- ii** Find the value of y when $x = \frac{\pi}{4}$.
- iii** Find the period of the function.
- iv** Find the distance between the asymptotes of the function.
- v** State the equation of the first asymptote of the function for $x \geq 0$.
- vi** State the equation of the first asymptote of the function for $x \leq 0$.
- vii** Sketch a graph of the function.

a $y = \tan x - 2$

b $y = 5 \tan x + 3$

35 Consider the given graph of a tangent function:



- a** State the transformation required to obtain the given function from $y = \tan x$
- b** State the equation of the graphed function.

